Chapter 2. Logic

2.1. Statements

In this chapter, we shall be mainly concerned with *statements*. A *statement* is a sentence of which it is meaningful to say whether its content is true or false.

Example 2.1 Each of the following sentences is a statement:

1. The number 3 is even.
2. $9 > 5$.
3. The product of two odd numbers must be odd.
4. Every student in this course is over 18.
5. The 85th digit in the decimal expansion of π is 3.
6. If 2 is greater than 3, then the sun rises from the west.

None of the following sentences can be regarded as a statement in the above sense:

7. The number 4 is stupid.
8. Please come to my office today.
9. Will the teacher know if I fall asleep while watching this video?

Every statement has a *truth value*, namely **True** (T) or **False** (F), but not both. Sometimes it may not be immediately clear whether a given statement is true or false. Sentence 5 above is an example. In fact, for a sentence to be a statement, it is not required that we are able to determine its truth value.

A sentence whose truth value depends on the input of certain variable is not a statement but an *open statement*.

Example 2.2 Consider the sentence

$$P(x) : 4x = 16.$$

Then it is clear that the truth value of $P(x)$ depends on x. It is true when $x = 4$, that is, $P(4)$ is a true statement; otherwise it is false, for example, $P(1)$, $P(\pi)$ are false statements. □

The possible truth values of a statement are often given in a table, called a *truth table*. For two statements P and Q, there are four possible combinations of truth values for P and Q:

P	Q
T	T
T	F
F	T
F	F

For three statements P, Q and R, there are eight possible combinations of truth values for P, Q and R:

P	Q	R
T	T	T
T	T	F
T	F	T
T	F	F
F	T	T
F	T	F
F	F	T
F	F	F

In general, a truth table involving n statements contains 2^n possible combinations of truth values for these statements.

2.2. Conjunctions, Disjunctions and Negations

We now consider different modes of combining statements:

(i) Conjunctions

The *conjunction* of two statements P and Q, denoted by $P \wedge Q$, is a statement which is true if both P and Q are true; otherwise it is false. Thus the truth value of a conjunction $P \wedge Q$ is uniquely determined by the truth values of its components P and Q. For example, let P and Q be statement 1 and 2 of Example 2.1 respectively, then their conjunction $P \wedge Q$ is a new statement

The number 3 is even *and* $9 > 5$,

which is false since P is false (even Q is true).

The truth table for $P \wedge Q$ is as follows:

P	Q	$P \wedge Q$
T	T	T
T	F	F
F	T	F
F	F	F

(ii) Disjunctions

The *disjunction* of two statements P and Q, denoted by $P \vee Q$, is a statement which is false if both P and Q are false; otherwise it is true. For example, using the statements P and Q as before, we can form a new statement $P \vee Q$

The number 3 is even *or* $9 > 5$,

which is true since Q is true (even P is false).

The truth table for $P \vee Q$ is as follows:

P	Q	$P \vee Q$
T	T	T
T	F	T
F	T	T
F	F	F

Remark: The use of the word "or" in mathematics may be different from its common usage as an English word. For example when someone says "I will stay at home or go to school this afternoon", it is common to interpret this statement as "either I will stay at home, or I will go to school, but not both". The usage of "or" in this case is interpreted in the exclusive sense, meaning 'one or the other but not both'. However in mathematics, the word "or" is always used inclusively. Thus the statement $P \vee Q$ is also true when both P and Q are true simultaneously, as seen from the truth table above.

(iii) Negations

Each of the conjunction and disjunction combines two statements into a new statement. The *negation* transforms a single given statement to a new one. The *negation* of a statement P, denoted by $\sim P$, is a statement which has the opposite truth value of P. The truth table for $\sim P$ is as follows:

P	$\sim P$
T	F
F	T

The usual way to express a negation is to put the word *not* into a statement or to delete it from a statement. For example, using the statements P and Q as before, their negations are

$\sim P$: The number 3 is not even. (Or equivalently, The number 3 is odd.)

and

$$\sim Q : \ 9 \leq 5.$$

Exercise Write down the negation of the statement "Some students in HKU do not live in a hall."

2.3. Conditionals

The "*if-then*" combination provides another way of forming a new statement. The *conditional* of two statements P and Q is the statement

$$\text{If } P \text{ then } Q$$

and is denoted by $P \to Q$. Here P is called the *hypothesis* and Q is called the *consequent* of the conditional.

The conditional $P \to Q$ is false when P is true and Q is false; otherwise it is true. The truth table for $P \to Q$ is as follows:

P	Q	$P \to Q$
T	T	T
T	F	F
F	T	T
F	F	T

For example, using the statements P and Q as before, the conditional is the statement

$$\text{If the number } 3 \text{ is even, then } 9 > 5.$$

This is a conditional with a false hypothesis and a true consequent. Therefore it is a true statement.

Example 2.3 Consider the statement:

$$\text{If you get an } A \text{ for MATH2012, then I buy you a cheesecake.}$$

The above statement is true either if you really get an A for MATH2012 and I then buy you a cheesecake, or you do not get an A for MATH2012 (in this case, it does not matter whether I buy you a cheesecake or not). The only situation that this statement becomes false is that, if you got an A for MATH2012, but I did not buy you a cheesecake (that is, I have broken my promise!) □

The Biconditional

The statement $Q \to P$ is called the *converse* of the statement $P \to Q$. In mathematics, when we are given that $P \to Q$ is true, it is very common to question whether its converse is also true, that is, whether $Q \to P$ is true.

For example, the converse of the statement in Example 2.3 is: If I buy you a cheesecake, then you get an A for MATH2012.

The *biconditional* of two statements P and Q, denoted by $P \leftrightarrow Q$, is the conjunction

$$(P \to Q) \wedge (Q \to P)$$

of the conditional and its converse. This means

$$\text{"If } P \text{ then } Q\text{"} \quad \text{and} \quad \text{"If } Q \text{ then } P\text{"}$$

or in short, "P if and only if Q", or simply "P iff Q". The truth table for $P \leftrightarrow Q$ is as follows:

P	Q	$P \to Q$	$Q \to P$	$(P \to Q) \wedge (Q \to P)$
T	T	T	T	T
T	F	F	T	F
F	T	T	F	F
F	F	T	T	T

or in short,

P	Q	$P \leftrightarrow Q$
T	T	T
T	F	F
F	T	F
F	F	T

Observe that the biconditional $P \leftrightarrow Q$ is true when both P and Q have the same truth values; otherwise it is false.

2.4. Implication and Logical Equivalence

Implication

The conditional $P \to Q$ is a statement. It may have truth value either T or F. If the conditional is true, then we can say "P implies Q". Symbolically, we use $P \Rightarrow Q$ to denote the *implication* "P implies Q". That is to say,

$$P \Rightarrow Q \quad \text{means} \quad \text{the statement "}P \to Q\text{" is true.}$$

Venn Diagram

There are two other equivalent ways of saying "P implies Q":

- "P is a *sufficient condition* for Q"; (or in short, P is *sufficient* for Q)
- "Q is a *necessary condition* for P" (or in short, Q is *necessary* for P)

Example 2.4 Let P and Q be statements. Show that $P \wedge Q \Rightarrow P$.

Solution. To show that the above is an implication, we need to show that the conditional $P \wedge Q \to P$ is always true, regardless of the truth values of P and Q. This can be seen from the following truth table:

P	Q	$P \wedge Q$	$P \wedge Q \to P$
T	T	T	T
T	F	F	T
F	T	F	T
F	F	F	T

□

Example 2.5 Consider the open statements

$$P(x) : x = -10 \quad \text{and} \quad Q(x) : |x| = 10.$$

Show that we have the implication $P(x) \Rightarrow Q(x)$ for any real number x.

Solution. If $x = -10$, then both $P(-10)$ and $Q(-10)$ are true. Hence $P(-10) \to Q(-10)$ is true. If $x \neq -10$, then $P(x)$ is false. But $|x|$ can be 10 or not equal to 10, that is, $Q(x)$ can be true or false. In any case, $P(x) \to Q(x)$ is true for $x \neq -10$ (since the hypothesis is false). Combining with the case $x = -10$, we have $P(x) \to Q(x)$ is true for any real number x. In other words, $P(x) \Rightarrow Q(x)$ for any real number x. □

Suppose we know that $P \Rightarrow Q$. In many situation we are interested in whether the converse is true or not, that is, whether we have $Q \Rightarrow P$. In case we have $P \Rightarrow Q$ but $Q \not\Rightarrow P$, then we say that

1. P is *sufficient but not necessary for* Q (or equivalently, Q *is necessary but not sufficient for* P)
2. P is a *stronger condition* (or equivalently, Q is a *weaker condition*)
3. P is *stronger than* Q (or equivalently, Q is *weaker than* P)

When it is the case that we have both $P \Rightarrow Q$ and $Q \Rightarrow P$, then we say that "P *is logically equivalent to* Q". Logical equivalence will be discussed in the following section.

Logical Equivalence

If the biconditional "$P \leftrightarrow Q$" is true, then we shall say that "P is *logically equivalent to* Q", written as $P \Leftrightarrow Q$. That is to say,

$$P \Leftrightarrow Q \quad \text{means} \quad \text{the statement "}P \leftrightarrow Q\text{" is true.}$$

In other words, two statements are logically equivalent if they have the same truth values. Other equivalent ways of saying "P is logically equivalent to Q" include: "P if and only if Q" or "P is *necessary and sufficient* for Q".

Example 2.6 Consider the open statements

$$P(x) : x = 10 \text{ or } x = -10 \qquad \text{and} \qquad Q(x) : |x| = 10.$$

Show that we have $P(x) \Leftrightarrow Q(x)$ for any real number x.

Solution. If $x = 10$, then both $P(10)$ and $Q(10)$ are true. Hence $P(10) \leftrightarrow Q(10)$ is true. The same situation holds when $x = -10$, that is, $P(-10) \leftrightarrow Q(-10)$ is true. On the other hand, if x is not equal to 10 or -10, then $P(x)$ is false and in this case $|x|$ cannot be 10, that is, $Q(x)$ is false. Hence both $P(x)$ and $Q(x)$ are false for any real number x other than 10 or -10. Thus $P(x) \leftrightarrow Q(x)$ is true for any real number x other than 10 or -10. It follows that $P(x) \Leftrightarrow Q(x)$ for any real number x. □

Example 2.7 Each of the following pair of statements are logically equivalent:

(i) $\sim(\sim P)$ and P (double negation)

(ii) $P \wedge Q$ and $Q \wedge P$ (commutative law of conjunction)

(iii) $P \vee Q$ and $Q \vee P$ (commutative law of disjunction)

(iv) $P \wedge (Q \wedge R)$ and $(P \wedge Q) \wedge R$ (associative law of conjunction)

(v) $P \vee (Q \vee R)$ and $(P \vee Q) \vee R$ (associative law of disjunction)

(vi) $P \vee (Q \wedge R)$ and $(P \vee Q) \wedge (P \vee R)$ (distributive law)

(vii) $P \wedge (Q \vee R)$ and $(P \wedge Q) \vee (P \wedge R)$ (distributive law)

(viii) $\sim(P \wedge Q)$ and $(\sim P) \vee (\sim Q)$ (De Morgan's law)

(ix) $\sim(P \vee Q)$ and $(\sim P) \wedge (\sim Q)$ (De Morgan's law)

Let us verify (v), (vi) and (viii); others are left as exercise. The truth table for the two statements in (v) is as follows:

If the truth table is same. The two conditions are same.

P	Q	R	$P \vee Q$	$Q \vee R$	$P \vee (Q \vee R)$	$(P \vee Q) \vee R$
T	T	T	T	T	T	T
T	T	F	T	T	T	T
T	F	T	T	T	T	T
T	F	F	T	F	T	T
F	T	T	T	T	T	T
F	T	F	T	T	T	T
F	F	T	F	T	T	T
F	F	F	F	F	F	F

For (vi):

P	Q	R	$Q \wedge R$	$P \vee Q$	$P \vee R$	$P \vee (Q \wedge R)$	$(P \vee Q) \wedge (P \vee R)$
T	T	T	T	T	T	T	T
T	T	F	F	T	T	T	T
T	F	T	F	T	T	T	T
T	F	F	F	T	T	T	T
F	T	T	T	T	T	T	T
F	T	F	F	T	F	F	F
F	F	T	F	F	T	F	F
F	F	F	F	F	F	F	F

For (viii):

P	Q	$\sim P$	$\sim Q$	$P \wedge Q$	$\sim (P \wedge Q)$	$(\sim P) \vee (\sim Q)$
T	T	F	F	T	F	F
T	F	F	T	F	T	T
F	T	T	F	F	T	T
F	F	T	T	F	T	T

Example 2.8 Show that the statements $P \to Q$ and $(\sim P) \vee Q$ are logically equivalent.

Solution. The truth table below suggests their equivalence:

P	Q	$\sim P$	$P \to Q$	$(\sim P) \vee Q$
T	T	F	T	T
T	F	F	F	F
F	T	T	T	T
F	F	T	T	T

□

Example 2.9 Show that the statements $P \to Q$ and $(\sim Q) \to (\sim P)$ are logically equivalent.

Solution. Instead of using the truth table, we are going to apply the laws from Example 2.7 and 2.8 to show the equivalence:

$$P \to Q \iff (\sim P) \vee Q \iff Q \vee (\sim P) \iff (\sim (\sim Q)) \vee (\sim P) \iff (\sim Q) \to (\sim P) \quad \square$$

Remark The statement $(\sim Q) \to (\sim P)$ is known as the *contrapositive* of the conditional $P \to Q$. Example 2.9 tells us that the conditional and its contrapositive are logically equivalent. The fact that they are equivalent will be essential when we discuss *Proof by Contrapositive* later.

Tautologies and Contradictions

Let P be any statement. The statement $P \vee (\sim P)$ is always true regardless of the truth value of P. Such statement is known as a *tautology* and we denote it by **T**. Another example of a tautology is given by $(\sim Q) \vee (P \to Q)$ (verify it!)

On the other hand, the statement $P \wedge (\sim P)$ is always false regardless of the truth value of P. Such statement is known as a *contradiction* and we denote it by **F**. Another example of a contradiction is given by $(P \wedge Q) \wedge (Q \to (\sim P))$ (verify it!)

The following are some laws involving tautologies and contradictions:

$$P \vee \mathbf{T} \Leftrightarrow \mathbf{T}, \quad P \wedge \mathbf{T} \Leftrightarrow P, \quad P \vee \mathbf{F} \Leftrightarrow P, \quad P \wedge \mathbf{F} \Leftrightarrow \mathbf{F}, \quad (\sim \mathbf{T}) \Leftrightarrow \mathbf{F}, \quad (\sim \mathbf{F}) \Leftrightarrow \mathbf{T}$$

Exercise Apply the laws from Example 2.7, 2.8, 2.9, tautologies and contradictions, prove the following equivalence of statements:

(i) $\sim (P \to Q) \iff P \wedge (\sim Q)$

(ii) $P \wedge (P \to Q) \iff P \wedge Q$

(iii) $(P \wedge Q) \to P \iff \mathbf{T}$

(iv) $(\sim P) \to (Q \wedge (\sim Q)) \iff P$

(v) $P \wedge \{[R \to (P \vee Q)] \leftrightarrow [(R \to P) \vee (R \to Q)]\} \iff P$

2.5. Quantifiers

Consider the statement "The square of every real number is nonnegative". This statement can be rephrased as "For every real number x, we have $x^2 \geq 0$". Each of the phrases

"for every", "for each", "for all", "for any", "whenever"

is referred to as the *universal quantifier* and is commonly denoted by the symbol $\forall$. The above statement can then be written as

$$\forall x \in \mathbb{R},\ x^2 \geq 0.$$

If we write $P(x)$ as the open statement $P(x) : x^2 \geq 0$, then we can also write the above as $\forall x \in \mathbb{R},\ P(x)$. This means that for any x in the set of real numbers (the universal set in this example), the property $P(x)$ holds.

Now consider another statement "There exists an integer x such that $x^2 = 2$". Each of the phrases

"there exists", "there is", "for some", "for at least one"

is referred to as the *existential quantifier* and is commonly denoted by the symbol $\exists$. Hence we can write the above statement as

$$\exists x \in \mathbb{Z} \text{ such that } x^2 = 2 \quad \text{or} \quad \exists x \in \mathbb{Z},\ x^2 = 2.$$

In summary, if $P(x)$ is any open statement, and if we are concerning the elements x in some universal set S, then we have

"$\forall x \in S,\ P(x)$" means "$P(x)$ is true for any $x \in S$",

and

"$\exists x \in S,\ P(x)$" means "There exists at least one $x \in S$ such that $P(x)$ is true."

Next we consider the situation when we need to find the negations of statements containing quantifiers. For example,

$$\sim (\forall x \in S,\ P(x)) \quad \text{means} \quad \text{"It is not the case that } P(x) \text{ is true for all } x \in S\text{."}$$

This means $P(x)$ is false for at least one $x \in S$, that is, $\exists x \in S,\ \sim P(x)$. Likewise, $\sim (\exists x \in S,\ P(x))$ means $P(x)$ is false for all $x \in S$, that is, $\forall x \in S,\ \sim P(x)$. To summarize, we have

$$\sim (\forall x \in S,\ P(x)) \quad \text{is logically equivalent to} \quad \exists x \in S,\ \sim P(x),$$

$$\sim (\exists x \in S,\ P(x)) \quad \text{is logically equivalent to} \quad \forall x \in S,\ \sim P(x).$$

Example 2.10 Consider the open statement $P(x) : x$ is an even integer. Then

$$\text{"} \sim (\forall x \in \mathbb{Z},\ P(x))\text{"} \quad \text{means} \quad \text{"Not every integer is even."}$$

It is logically equivalent to

$$\text{"}\exists x \in \mathbb{Z},\ \sim P(x)\text{"} \quad \text{which means} \quad \text{"Some integer is not even."}$$

Moreover,

$$\text{"} \sim (\exists x \in \mathbb{Z},\ P(x))\text{"} \quad \text{means} \quad \text{"There is no even integer."}$$

It is logically equivalent to

$$\text{"}\forall x \in \mathbb{Z},\ \sim P(x)\text{"} \quad \text{which means} \quad \text{"Every integer is not even."}$$ □

A statement may contain two (or more) quantifiers. For example, "For any integer x and for any integer y, the sum $x + y$ is an integer." can be written as

$$\forall x \in \mathbb{Z},\ \forall y \in \mathbb{Z},\ x + y \in \mathbb{Z}.$$

It is understood that interchanging $\forall x \in \mathbb{Z}$ and $\forall y \in \mathbb{Z}$ will not affect the meaning of the statement. Hence we can also write the above statement as

$$\forall x,\, y \in \mathbb{Z},\ x + y \in \mathbb{Z}.$$

Here is an example involving two existential quantifiers: "There exists a real number x, there exists a real number y such that $xy = 1$." can be written as

$$\exists x \in \mathbb{R},\ \exists y \in \mathbb{R},\ xy = 1.$$

Similarly, interchanging $\exists x \in \mathbb{R}$ and $\exists y \in \mathbb{R}$ will not affect the meaning of the statement. Hence we can also write the above statement as

$$\exists x,\, y \in \mathbb{R},\ xy = 1.$$

The two quantifiers $\forall$ and $\exists$ can also be used together. For example, "For any integer x, there exists an integer y such that $y > x$." This can be written as

$$\forall x \in \mathbb{Z},\ \exists y \in \mathbb{Z},\ y > x.$$

It says that for each integer, we can find a greater one. This is certainly a true statement. For example, if you give me $x = 27$, then I can take $y = 30$, 100, or 2415. Note that, unlike the previous examples, **the order of the two quantifiers is important to the meaning of the statement**. Reversing the quantifiers gives

$$\exists y \in \mathbb{Z},\ \forall x \in \mathbb{Z},\ y > x.$$

This statement is indeed false. It says that I can find an integer y so that if you give me an integer x, then my y is greater than your x. In other words, I can find an integer which is greater than all integers (including itself!) Thus we see that **interchanging the positions of $\forall$ and $\exists$ will change the meaning of the statement**. Here is another example:

(i) In every course in HKU, there is a student who fails the final examination.

(ii) There is a student who in every course in HKU fails the final examination.

Now consider the two statements again:

$$\forall x \in \mathbb{Z},\ \exists y \in \mathbb{Z},\ y > x$$

and

$$\exists y \in \mathbb{Z},\ \forall x \in \mathbb{Z},\ y > x.$$

In the first statement, the expression $\exists y \in \mathbb{Z}$ appears *after* the introduction of the variable x. It therefore follows from the structure of the sentence that the variable y may depend on x.

In the second statement, on the other hand, y cannot possibly depend on x, because x has not even been introduced when y is supposed to be determined. Thus we have another principle: **a variable associated to a quantifier $\exists$ may (or may not) depend on variables which have been introduced earlier, but must be independent of any variables introduced later in the statement**. Using the examples taken from (i) and (ii) above, the student depends on the course (introduced earlier) in (i) but not on the course (introduced later in the statement) in (ii). Here is another example (make sure you have the correct interpretation!):

(iii) For every university in HK, there is a compulsory course in mathematics for all students in that university.

(iv) For every university in HK and for all students in that university, there is a compulsory course in mathematics.

2.6. Some Applications

Besides using them to count, sets can also be used to represent logical implications. For example, let C be the set of all good cats and M be the set of animals that catch mice. Then the statement "All good cats catch mice" can be written as $\forall x \in C,\ x \in M$, that is, $C \subseteq M$. With this interpretation, sets can be used to deduce logical conclusions. For example, if $C \subseteq M$, then $M^c \subseteq C^c$ (here we take the set of all animals as the universal set). Hence we can conclude that "An animal that can't catch mice cannot be a good cat".

Example 2.11 Consider the following statements:

(a) Irrational people can never be smart.

(b) Rational people are always careful.

(c) Great men are always energetic.

(d) Energetic people are always careless.

(e) Artful people are always smart.

Suppose all statements above are true. Let S, R, E, C, G, A be the set of all people who are Smart, Rational, Energetic, Careless, Great and Artful respectively. Then the five statements above are equivalent to

(a) $R^c \subseteq S^c$.

(b) $R \subseteq C^c$, or equivalently $C \subseteq R^c$.

(c) $G \subseteq E$.

(d) $E \subseteq C$.

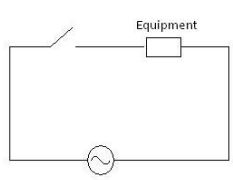
(e) $A \subseteq S$, or equivalently $S^c \subseteq A^c$.

Thus we have

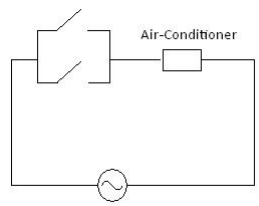
$$G \subseteq E \subseteq C \subseteq R^c \subseteq S^c \subseteq A^c.$$

We conclude that great man is not artful. In particular, we can draw the conclusion that $G \subseteq S^c$, that is, "all great men are not smart", or equivalently, "smart people can never be great", as we also have $S \subseteq G^c$. □

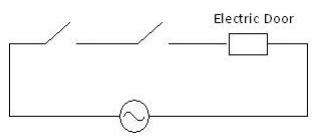
Another area of application concerns switching circuits. As shown in the diagram below, the position of a single switch allows us to control the on/off of an equipment.



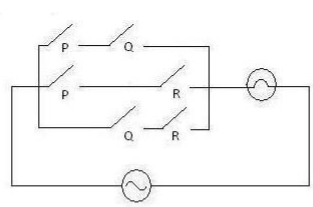
In many circumstances, more complicated systems of switches are needed in order to achieve satisfactory control of some equipments. For example, an air-conditioning system might be controlled by the thermostats in two rooms. If the temperature in either room exceeds $25°$C, then the thermostat in that room will close a switch to turn on the air-conditioning system. In such case, the switches should be interconnected in *parallel* as shown below:



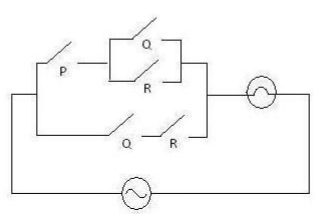
On the other hand, the door to the vault of a bank might be controlled electronically so that it cannot be opened unless two separate keys are used to close two electric switches simultaneously. That is, the switches are inter-connected in *series*, as shown below:



Clearly, when switches are in parallel, we achieve a closed circuit when one or both switches are closed. Likewise, when switches are in series, a closed circuit is achieved when both switches are closed. More complicated control pattern can be achieved by combining parallel and series switches suitably. For example, a voting machine for three people P, Q and R can be designed in the following way: The statement we want is $(P \wedge Q) \vee (P \wedge R) \vee (Q \wedge R)$ (win by simple majority), which is realized by the following circuit:



But, $(P \wedge Q) \vee (P \wedge R) \vee (Q \wedge R) \Longleftrightarrow (P \wedge (Q \vee R)) \vee (Q \wedge R)$. An equivalent circuit is given by



which uses fewer switches.